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Vector Calculus - GATE Study Material in PDF

In previous articles, we have already seen the basics of Calculus – Differentiation and Integration and applications. In **GATE 2018** Study Notes, we will be introduced to **Vector Calculus**. A vector has both magnitude and direction whereas a scalar has only magnitude. Let us now see how to perform certain operations on vectors.

These GATE study material are useful for GATE EC, GATE EE, GATE ME, GATE CS, GATE CE and all other branches. Also useful for exams such as BARC, BSNL, DRDO, IES, ISRO, ECIL etc. You can have these notes downloaded as PDF so that your exam preparation is made easy and you ace your paper. Before you get started, go through the basics of Engineering Mathematics.

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Recommended Reading –

Types of Matrices

Properties of Matrices

Rank of a Matrix & Its Properties

Solution of a System of Linear Equations

Eigen Values & Eigen Vectors

Linear Algebra Revision Test 1

Laplace Transforms

Limits, Continuity & Differentiability

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Mean Value Theorems

Differentiation

Partial Differentiation

Maxima and Minima

Methods of Integration & Standard Integrals

Dot Product

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are two constant vectors.

Then Dot Product of two vectors is given by

a. $b = |a||b| \cos\theta$ where θ = angle between $\vec{a}, \vec{b}$.

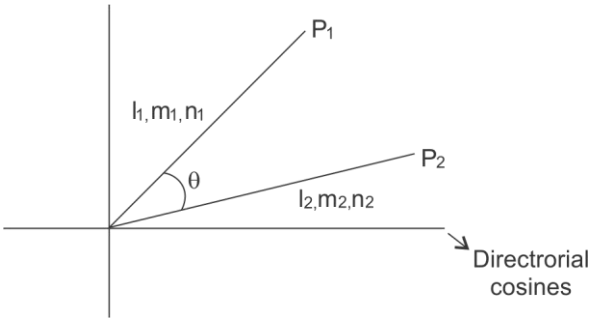
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Note:

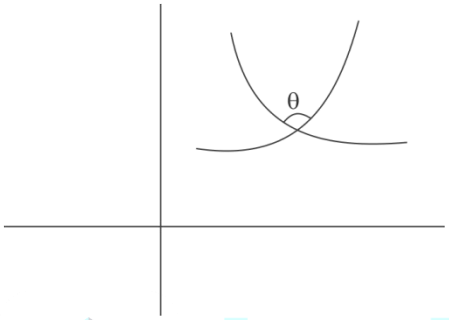
1. $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \quad (\because \theta = 0^0)$
2. $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0 \ (\because \theta = 90^0)$
3. Magnitude of Vector $\vec{a} = |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
4. Angle between two vectors $\theta = \cos^{-1} \frac{\vec{a} \cdot \vec{b}}{|a||b|}$
5. Also, $\cos\theta = \frac{l_1 l_2 + m_1 m_2 + n_1 n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{l_2^2 + m_2^2 + n_2^2}}$

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6. If slopes are given and angle between two curves is θ then $\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$



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7. If $\vec{a} \cdot \vec{b} = 0 \Rightarrow$ Vectors are orthogonal ($\theta = 90°$)

8. If $\vec{a} \cdot \vec{b} = |a||b| \Rightarrow$ Vectors are parallel ($\theta = 0°$)

Cross Product

If $\bar{a}$ and $\bar{b}$ are two vectors then then cross product between two vectors is given by

$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\, \hat{n}$ where $\hat{n} =$ unit vector normal to both $\vec{a}$ and $\vec{b}$

Note:

1. $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0 \quad (\because \theta = 0^0)$




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2. $\hat{i} \times \hat{j} = \hat{j} \times \hat{k} = \hat{k} \times \hat{i} = 1 \quad (\because \theta = 90^0)$

3. Angle between two vectors is θ then $\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|a||b|}$

4. If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then cross product between two vectors

can be calculated as $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

$= \hat{i}(a_2b_3 - a_3b_2) - \hat{j}(a_1b_3 - a_3b_1) + \hat{k}(a_1b_2 - a_2b_1)$

5. Geometrically cross product gives the area of triangle

6. If $\overrightarrow{AB}, \overrightarrow{AC}$ are two sides of triangle then area of the triangle is $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$

Triple Product

1. Geometrically Triple Product gives the Volume of Tetrahedron

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = [a\ b\ c] = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

2. If $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0 \Rightarrow$ Vector are coplanar vectors

Derivative of a Vector

Let $\vec{r} = f(t)$ be a position vector where t is a scalar variable.

$\vec{r} + \delta\vec{r} = f(t + \delta t)$

$\delta\vec{r} = f(t + \delta(t) - f(t)$

$\lim_{\delta t \to 0} \frac{\delta\vec{r}}{\delta t} = \frac{f(t+\delta(t)-f(t)}{\delta t}$ is called $\frac{d\vec{r}}{dt}$

$\therefore \frac{d\vec{r}}{dx} = \lim_{\delta t \to 0} f\frac{(t+\delta t)-f(t)}{\delta t}$

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Vector differentiation is nothing but ordinary differentiation but only difference is $\vec{r}$ is position vector.

Formulae:

1. $\frac{d}{dt}(\vec{A} \pm \vec{B}) = \frac{d}{dt}\vec{A} \pm \frac{d}{dt}\vec{B}$
2. $\frac{d}{dt}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{dt} \pm \frac{d\vec{A}}{dt} \cdot \vec{B}$ Where $\vec{A} = A(t)$ and $\vec{B} = B(t)$
3. $\frac{d}{dt}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{d\vec{B}}{dt} \pm \frac{d\vec{A}}{dt} \times \vec{B}$

Vector Operator (∇ - Del)

$\vec{\nabla} = \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$ is called vector operator

Gradient

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If ϕ (x, y, z) be a given scalar function then $\vec{\nabla}\phi$ is called gradient.

$$\vec{\nabla}\phi = \hat{i}\frac{\partial \phi}{\partial x} + \hat{j}\frac{\partial \phi}{\partial y} + \hat{k}\frac{\partial \phi}{\partial z}$$

Note:

1. Physically, gradient gives rate of change of ϕ w.r.t x, y, z separately.
2. Geometrically, it gives normal to the level surface.

Example 1:

If $\phi = xyz$ then find the value of $\vec{\nabla}\phi$.

Solution:

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$\vec{\nabla}\phi = \hat{i}\frac{\partial}{\partial x}(xyz) + \hat{j}\frac{\partial}{\partial y}(xyz) + \hat{k}\frac{\partial}{\partial z}(xyz)$

$\vec{\nabla}\phi = \hat{i}yz + \hat{j}xz + \hat{k}xy$

Note:

1. $\vec{\nabla}(\log r) = \frac{\vec{r}}{r^2}$

2. $\vec{\nabla}(\sin r) = \cos r \cdot \frac{\vec{r}}{|r|}$

3. $\vec{\nabla}(r^n) = n \cdot r^{n-1} \cdot \vec{\nabla}r = n \cdot r^{n-1} \cdot \frac{\vec{r}}{|r|} = n \cdot r^{n-2}\vec{r}$

4. Let $\phi(x,y,z) = c$ be given equation of the level surface then the outward unique normal vector is defined as $\vec{N} = \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|} = \frac{\text{Grad}(\phi)}{|\text{Grad}(\phi)|}$

Example 2:

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Find the value of unit normal vector $\hat{N}$ for the sphere $x^2 + y^2 + z^2 = 9$.

Solution:

$\phi = x^2 + y^2 + z^2 - 9$

$\vec{\nabla}\phi = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z)$

$|\vec{\nabla}\phi| = 2\sqrt{x^2 + y^2 + z^2}$

$\hat{N} = \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{\sqrt{x^2+y^2+z^2}} = \frac{x\hat{i}+y\hat{j}+z\hat{k}}{3}$

Angle between Two Surfaces

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Let $\phi_1(x,y,z) = C$, $\phi_2(x,y,z) = C$ be given equations of two level surfaces and angle between these two surfaces are given as θ then $\cos\theta = \frac{\nabla\phi_1 . \nabla\phi_2}{|\nabla\phi_1||\nabla_2|}$

Note:

The angle between two surfaces is nothing but the angle between their normal.

$\vec{\nabla}\phi_1 . \vec{\nabla}\phi_2 = 0 \Rightarrow$ then they are said to be orthogonal surfaces

Example 3:

The angle between the two surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$ is

Solution:

Here $\phi_1 = x^2 + y^2 + z^2 - 9$

$\vec{\nabla}\phi_1 = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z) \Rightarrow |\vec{\nabla}\phi_1| = 2\sqrt{x^2 + y^2 + z^2} = 2\sqrt{9} = 6$

$\phi_2 = x^2 + y^2 - z - 3$

$\vec{\nabla}\phi_2 = \hat{i}(2x) + \hat{j}(2y) - \hat{k} \Rightarrow |\vec{\nabla}\phi_2| = \sqrt{4x^2 + 4y^2 + 1} = \sqrt{21}$

$\cos\theta = \frac{\vec{\nabla}\phi_1 . \vec{\nabla}\phi_2}{|\vec{\nabla}\phi_1||\vec{\nabla}\phi_2|} = \frac{(4x^2+4y^2-2z)}{6\sqrt{21}} = \frac{16+4-4}{6\sqrt{21}} = \frac{8}{3\sqrt{21}}$

Directional Derivatives of a Scalar Function

The directional derivative of a scalar function $\phi\,(x, y, z)$ in the direction of a vector $\bar{a}$ is given as $\vec{\nabla}\phi \cdot \hat{e}$ where $\hat{e} = \frac{\vec{a}}{|\vec{a}|}$

If $\vec{\nabla}\phi \cdot \hat{e}$ is – ve then it is in the opposite direction.

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Example 4:

The Directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, -1)$along $(2i - j - 2k)$ is

Solution:

$\vec{\nabla}\phi = \hat{i}[(2xyz) + 4z^2] + \hat{j}[x^2z] + \hat{k}[x^2y + 8xz]$

Directional Derivative $= \vec{\nabla}\phi \cdot \hat{e}$

At $(1, -2, -1)$we have $\vec{\nabla}\phi \cdot \hat{e} = [4 + 4]\hat{i} + (-\hat{j}) + \hat{k}[-2 - 8] \cdot \frac{(2\hat{i} - \hat{j} - 2\hat{k})}{3}$

$= \frac{16}{3} + \frac{1}{3} + \frac{20}{3} = \frac{37}{3}$

Divergence of a Vector

If $\vec{F}$ is a vector point function then $\vec{\nabla} \cdot \vec{F}$ is called Divergence of $\vec{F}$

Where $\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ and F_1, F_2, F_3 are the functions of x, y, z

$\therefore \vec{\nabla} \cdot \vec{F} = \left[\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right] \cdot \left[F_1\hat{i} + F_2\hat{j} + F_3\hat{k}\right]$

$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

Note:

1. Divergence of a vector is scalar.

2. Physically Divergence measures (outflow - inflow)

3. A vector whose divergence is zero then it is said to be divergence free vector (or) solenoid vector i.e. outflow = inflow = constant

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4. Geometrically, Divergence gives the rate at which the fluid entering in a rectangular parallelepiped per unit volume at the point.

Curl of a Vector

$\vec{\nabla} \times \vec{F}$ is called the curl of a vector where $\vec{F} = F_1\,\hat{\imath} + F_2\,\hat{\jmath} + F_3\hat{k}$

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \hat{\imath}\left(\frac{\partial F_3}{\partial x} - \frac{\partial F_2}{\partial z}\right) - \hat{\jmath}\left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right) + \hat{k}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

Note:

1. If $\vec{\nabla} \times \vec{F} = 0$ then it is said to be irrational vector otherwise it is said to be rotational vector.
2. Physically Curl gives the angular Velocity $\vec{w} = \frac{1}{2}(\vec{\nabla} \times \vec{V})$
3. Divergence of a curl of any vector is always zero.
4. $\vec{\nabla}.(\vec{\nabla}\phi) = \vec{\nabla}^2\phi$
5. $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$
6. $\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \vec{\nabla}^2\vec{F}$
7. $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A}.\vec{B})\vec{C}$ is known as a vector triple product.

Example 5:

The values of a, b, c so that the vector,

$\vec{V} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$ is irrotational

Solution:

Given, that vector V is irrotational


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$\therefore \vec{\nabla} \times \vec{V} = 0$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x + 2y + az) & (bx - 3y - z) & (4x + cy + 2z) \end{vmatrix}$$

$\Rightarrow \hat{i}(c + 1) + \hat{j}(4 - a) + \hat{k}(b - 2) = 0$

$\Rightarrow \ c = -1, a = 4, b = 2$

$\therefore \ a = 4, b = 2, c = -1$

So far we have seen about basic terminology in vector calculus and in the next article we will discuss about integration in vectors. Also we will discuss some important theorems which will convert one form of integral into another form of integral.

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